

Real-World Applications of Integers and Opposites

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Positive numbers are used to describe movement or situations in one direction, such as moving **up, forward, north, east**, earning money, or temperatures above zero.

Negative Numbers

Negative numbers are used to describe the opposite direction or situation, such as moving **down**, **backward**, **south**, **west**, losing money, or temperatures below zero.

Opposite Quantities

A positive and a negative number with the same value represent **opposite quantities**.

$$+5 \quad \text{and} \quad -5$$

These are opposites – same distance from zero, different directions.

Positive or Negative?

Positive (+)	Negative (—)
Up	Down
Forward	Backward
North / East	South / West
Earning	Losing
Above zero	Below zero

Examples

Example 1

A plane flies at an altitude of 9 000 meters above sea level. Represent this using an integer.

+9 000

“Above sea level” means positive.

Example 2

A miner digs 35 meters below the ground. Represent this using an integer.

$$-35$$

“Below the ground” means negative.

Example 3

A student withdraws ₱500 from a savings account. Represent this using an integer.

$$-500$$

Withdrawing (losing money) means negative.

Example 4

A runner moves 30 meters north. If south is negative, represent this using an integer.

$$+30$$

Since south is negative, north (the opposite direction) is positive.

Practice Exercises

Practice 1

Write a positive or negative number to represent a submarine that is 250 meters below sea level.

Practice 2

A hiker climbs 180 meters above the starting point. Represent the situation using an integer.

Practice 3

The temperature is 9°C below zero. Write the corresponding integer.

Practice 4

A diver moves 15 meters downward from the water's surface. Represent the movement with an integer.

Practice 5

A person deposits ₱800 into a savings account.
Represent the transaction using an integer.

Practice 6

A student walks 12 meters west from the school gate. If east is positive, represent the movement with an integer.

Practice 7

An elevator goes up 7 floors from the ground floor. Represent the movement using an integer.

Activities

Activity 1

Write a positive or negative number to represent a cave that is 180 meters below sea level.

Activity 2

A balloon rises 240 meters above the ground.
Represent the situation using an integer.

Activity 3

The temperature is 14°C below zero. Write the corresponding integer.

Activity 4

A diver descends 22 meters below the water's surface. Represent the movement with an integer.

Activity 5

A customer deposits ₱1 200 into a bank account. Represent the transaction using an integer.

Activity 6

A cyclist travels 20 meters west. If east is positive, represent the movement with an integer.

Activity 7

An elevator moves up 11 floors. Represent the movement using an integer.

Thank you!